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EDITORIAL



Using AI to write scholarly publications

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Artificial intelligence (AI) natural language processing (NLP) systems, such as OpenAI's generative pre-trained transformer (GPT) model (<https://openai.com>) or Meta's Galactica (<https://galactica.org/>) may soon be widely used in many forms of writing, including scientific and scholarly publications (Heaven 2022).¹ While computer programs (such as Microsoft WORD and Grammarly) have incorporated automated text-editing features (such as checking for spelling and grammar) for many years, these programs are not designed to create content. However, new and emerging NLP systems are, which raises important issues for research ethics and research integrity.²

NLP is a way of enabling computers to interact with human language. A key step in NLP, known as tokenization, involves converting unstructured text into structured text suitable for computation. For example, the sentence "The cat sat on the mat" can be structured by tagging its parts: "the [article] cat [noun] sat [verb, past tense] on [preposition] the [article] mat [noun]." Once the parts of the text have been tagged, they can be processed by means of algorithms designed to produce appropriate responses to text (i.e., language generation). Rudimentary NLP-systems, such as the first generation of chatbots that assisted customers on websites, operated according to thousands of human-written rules for processing and generating text.

Recent advances in computational speed and capacity and the development of machine-learning (ML) algorithms, such as neural networks, have led to tremendous breakthroughs in NLP (Mitchell 2020). Today's NLP systems use ML to produce and refine statistical models (with billions of parameters) for processing and generating natural language. NLP systems are trained on huge databases (45 terabytes or more) of text available on the internet or other sources. Initial training (or supervised learning) involves giving the system the text and then "rewarding" it for giving correct outputs, as determined by human trainers.³ Over time, NLP systems will reduce their percentage of erroneous outputs and will learn from the data (Mitchell 2020). While NLP systems continue to learn as they receive and process data beyond their initial training data, they do not "know" the meaning or truth-

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value of the text they receive, process, and generate. Their function is simply to generate understandable (i.e., grammatically correct) and appropriate (i.e., highly probable) text outputs in response to text inputs.

That said, NLP systems can generate coherent and informative text, ranging from a few sentences or paragraphs to an entire essay in response to specific prompts from the user, such as the topic, length, or writing style. For example, in response to the query, “Write a 300 word piece about the difference between research ethics and research integrity,” the OpenAI chatbot generated the following 356 words:

Research ethics and research integrity are two concepts that are closely related but not identical. Both are concerned with ensuring that research is conducted in a responsible and accountable manner, but they focus on different aspects of the research process.

Research ethics refers to the moral principles that guide the conduct of research. These principles include respect for human subjects, integrity in the research process, and the responsible use of research results. In other words, research ethics is concerned with the moral aspects of research and how it impacts those involved in the research and the broader community.

Research integrity, on the other hand, is concerned with the honesty and reliability of the research process. This includes ensuring that research is conducted in a transparent and unbiased manner, that data is accurately collected and analyzed, and that research findings are reported honestly and without distortion.

One key difference between research ethics and research integrity is that research ethics focuses on the moral principles that should guide research, whereas research integrity focuses on the specific practices and processes that ensure the reliability and honesty of research. Another difference is that research ethics is concerned with the impact of research on individuals and society, whereas research integrity is primarily concerned with the reliability and trustworthiness of the research itself.

Despite these differences, research ethics and research integrity are closely related and often overlap in practice. For example, the principle of respect for human subjects in research ethics is closely related to the need for transparency and honesty in the research process, which is a key aspect of research integrity. Similarly, the responsible use of research results, which is a key principle of research ethics, is closely tied to the reliability and trustworthiness of the research, which is an essential aspect of research integrity.

Overall, research ethics and research integrity are both important for ensuring that research is conducted in a responsible and accountable manner. While they focus on different aspects of the research process, they are closely related and must be considered together to ensure that research is conducted in a way that is ethical, reliable, and trustworthy

(OpenAI ChatGPT 2022a).

This response is remarkably good. However, the OpenAI chatbot can also produce answers that contain erroneous, misleading, or irrelevant information. For example, in response to the query: “Did Immanuel Kant believe that ought implies can?” the OpenAI chatbot wrote:

Immanuel Kant did not believe that ought implies can. In fact, he argued that the opposite was true: he believed that can implies ought. This means that, according to Kant, if someone is able to do something, then they have a moral obligation to do it. In other words, if someone has the ability to do something good or moral, then they ought to do it
(OpenAI ChatGPT 2022b).

The OpenAI chatbot gave a completely **incorrect answer** to this query, because Kant explicitly said that ought implies can (Kohl 2015). The converse of ought implies can, proposed by the OpenAI chatbot, could lead to disastrous outcomes. Just because I can cheat on an exam, steal money, or kill someone, does not mean that I should do these things.

In response to the query, “Why do some ethicists and researchers believe that there should be upper limits on the risks that healthy adults can be exposed to in research that does not provide them a direct medical benefit?” the OpenAI chatbot discussed how principles of beneficence and respect for persons apply to the ethics of research and then concluded:

Overall, some ethicists and researchers believe that there should be upper limits on the risks that healthy adults can be exposed to in research that does not provide them a direct medical benefit because they believe that these individuals should not be subjected to unnecessary or excessive risks without their informed consent
(OpenAI ChatGPT2022c).

The problem with this answer is not that it is incorrect but that **it is irrelevant** to the main issue, i.e., whether there should be ethical limits to risks that healthy, freely consenting volunteers can be exposed to in research. Some have argued that limits on risks can be justified by appealing to strong paternalism or the need to safeguard public trust in research (Resnik 2012).

NLP systems raise some very interesting philosophical problems: Are they intelligent, and what does this mean in terms of human intelligence? Can they think? Do they have moral agency? Furthermore, NLP systems might help researchers in rewriting manuscripts, which would be especially useful for non-native (English) speakers. However, these uses of NLP would challenge our current understanding of originality and/or the author’s intellectual contribution to the task of writing. These are important questions for philosophers, computer scientists, and sociologists of science to ponder, but we will not address them here. Our concerns in this editorial are more practical.

First, using NLP systems raises issues related to accuracy, bias, relevance, and reasoning. As illustrated by the examples described above, these systems

are impressive but can still make glaring mistakes (Heaven 2022). Galactica developers warn that their language models can “Hallucinate,” “are Frequency-Biased” and “are often Confident But Wrong” (Galactica 2022; Heaven 2022). These flaws could be due to the fact that NLP systems only deal with statistical relationships among words and not relationships between language and the external world, which can lead them to make errors related to facts and commonsense reasoning (AI Perspectives 2020). Another well-known problem with many AI/ML systems, including NLP systems, is the potential for bias, because AI systems will reflect biases in the data they are trained on (Lexalytics 2022). For example, AI systems trained on data that includes racial, gender, or other biases will generate outputs that reproduce or even amplify those biases. NLP systems are also not very good at solving some mathematics problems (Lametti 2022) or evaluating text for relevance and coherence, and they may inadvertently plagiarize (AI Content Dojo 2021; Venture Beat 2021).

While NLP systems are likely to become better at minimizing bias, doing math, making relevant connections between concepts, and avoiding plagiarism, they are likely to continue to make factual and commonsense reasoning mistakes because they do not (yet) have the type of cognition or perception needed to understand language and its relationship to the external physical, biological, and social world. NLP systems can perform well when working with text already created or curated by humans, but can perform (dangerously) poorly when they lack human-generated data related to a topic and try to piece together text from different sources. Thus, ***any section of a manuscript written by an NLP system should be checked by a domain expert for accuracy, bias, relevance, and reasoning.***

Second, use of NLP systems raises issues of accountability. ***If a section of a manuscript written by an NLP system contains errors or biases, coauthors need to be held accountable for its accuracy, cogency, and integrity.*** While it is tempting to assign blame to the NLP systems and/or their developers for textual inaccuracies and biases, we believe that authors are ultimately responsible for the text generated by NLP systems and must be held accountable for inaccuracies, fallacies, or any other problems in manuscripts. We take this position because 1) NLP systems respond to prompts provided by researchers and do not proactively generate text; 2) authors can juxtapose text generated by an NLP system with other text (e.g., their own writing) or simply revise or paraphrase the generated text; and 3) authors will take credit for the text in any case. Researchers who use these NLP systems to write text for their manuscripts must therefore check the text for factual and citation accuracy; bias; mathematical, logical, and commonsense reasoning; relevance; and originality. If NLP systems write in English and authors have limited English proficiency, someone who is fluent in English must help them spot mistakes. If an NLP system makes a mistake (of omission or commission),

authors need to take precautionary measures to correct it before it is published. Reviewers and editors can and should help out with catching mistakes, but they often do not have the time or resources to check every claim made in a manuscript.

Third, use of NLP systems raises issues of transparency in relation to requirements for authorship credit and contributions. Since participation in the writing process is a requirement for becoming an author according to guidelines adopted by most journals (Resnik et al. 2016), and widely used contributor roles taxonomies (e.g., CRediT) make clear distinctions between writing the first draft and revising it (Hosseini et al. 2022), use of NLP systems should be acknowledged in the text (e.g., methods section) and mentioned in the references section. Because NLP systems may be used in ways that may not be obvious to the reader, ***researchers should disclose their use of such systems and indicate which parts of the text were written or co-written by an NLP system.*** The issue here is similar to the ghost writing/contribution problem in scientific publications, except that we are not (yet) ready to say that AIs should be listed as authors on manuscripts when they make substantial contributions. Even so, transparency requires that contributions by NLP systems should be specifically disclosed so that the reader has an accurate understanding of the writing of the paper.

Fourth, use of NLP systems raises issues of data integrity for research that involves the analysis of text, such as surveys, interviews, or focus groups. It is possible to use NLP systems to fabricate transcripts of interviews or answers to open-ended questions. While it has always been possible for researchers to fabricate or falsify text, NLP systems make it much easier to do this, because they can generate narratives quickly from a few simple prompts. Since we trust that readers of *Accountability in Research* (AiR) understand that any form of data fabrication or falsification is unethical and is prohibited by the journal, we see no need to issue a separate policy on data fabrication or falsification related to the use of AI to write text, but we would still like to call attention to this issue and stress that ***researchers should not use NLP systems to fabricate empirical data or falsify existing data.***

Fifth, ethical issues are not restricted to NLP-generated text only. It is possible, even likely, that researchers may employ these systems to generate an initial literature survey, find references, or synthesize ideas related to their work (e.g., <https://elicit.org/>), and then revise these suggestions to disguise their use (thereby making the human input look more impressive) and to prevent them from being identified by systems that detect NLP-generated content. But just as plagiarism can involve the misappropriation or theft of words or ideas, NLP-generated ideas may also affect the integrity of publications. **When NLP assistance has impacted the content of a publication (even in the absence of direct use of NLP-generated text), this should be disclosed.**

Finally, the issues discussed here go far beyond the use of AI to write text and impact research more generally. For a couple of decades now, researchers have used statistics programs, such as SPSS, to analyze data, and graphics programs, such as Photoshop, to process digital images. Ethical problems related to the misuse of statistics programs and digital image manipulation are well-known and have unfortunately been the subject of numerous research misconduct investigations (Gardenier and Resnik 2002; Rossner and Yamada 2004; Cromey 2013; Shamoo and Resnik 2022). Many biomedical journals have developed guidelines for using computer programs to process digital images (see Cell Press 2022) and the International Committee of Medical Journal Editors (2023) recommends that authors disclose the use of statistical software. We think that all uses of computer programs that substantially impact the content of the manuscript should be disclosed, but we will limit our focus here to uses of programs for writing or editing text.

In light of the rapidly-evolving nature of NLPs and ethical concerns with its use in research, the Editors of *Accountability in Research* are planning to adopt a policy on the inclusion of text and ideas generated by such systems in submissions to the Journal. The general goals of the policy will be, at a minimum, to ensure transparency and accountability related to use of these systems, while also being practical and straightforward. A draft of such a policy, and an invitation for submissions about this draft policy and these systems in general appear below.

Draft policy

All authors submitting manuscripts to *Accountability in Research* must **disclose** and describe the use of any *NLP systems* in writing the manuscript text or generating ideas used in the text and **accept full responsibility** for the text's factual and citation accuracy; mathematical, logical, and commonsense reasoning; and originality.

"NLP systems" are those that generate new content. For example, software that checks for spelling or offers synonyms or grammar suggestions does not generate new content per se, but NLP systems that develop new phrases, sentences, paragraphs, or citations related to specific contexts can influence the meaning, accuracy, or originality of the text, and should be disclosed.

Disclosures can be made in the methods section AND among the references, as appropriate. Authors should specify: 1) *who* used the system, 2) *the time and date of the use*, 3) *the prompt(s) used to generate the text*, 4) *the sections(s) containing the text*; and/or 5) *ideas in the paper resulting from NLP use*. Additionally, the text generated by NLP systems should be submitted as supplementary material. While this topic is a moving target and it may not be possible to anticipate all possible violations, an example of such a disclosure in the methods section could be: "In writing this manuscript, M.H. used OpenAI Chatbot on 9th of December 2022 at 1:21pm CST. The following prompt was used to write the introduction section: 'Write a 300 word piece about the difference between research ethics and research

integrity.’ The generated text was copied verbatim and is submitted as supplementary material.”

Accountability in Research is issuing a [call for submissions](#) focusing on the intersection of ethics, research integrity and policy related to NLP systems. We also invite commentary, exploration, and suggestions for improvements to our own policy draft above.

We encourage the editors of other journals to consider adopting policies on the use of AI in research, given the rapid and unpredictable advances in this technology. In the future, use of AI in research may raise issues of authorship, but that day has not yet arrived because today’s computing systems do not have the type of cognition, perception, agency, and awareness to be recognized as persons with authorship rights and responsibilities.

Notes

1. Blanco-González, Cabezón, Seco-González, et al. (2022) have recently posted a preprint on arXiv that tests the ability of ChatGPT in writing a scientific paper. They describe how the AI program was used.
2. NLP systems also raise important issues for academic integrity in colleges and universities and K-12 education, but we will not consider those here. For more on this see Stokel-Walker (2022).
3. While discussing the ethics of employing trainers, and the NLP systems’ need for massive human and financial resources (for training and improvement purposes) are outside the scope of this editorial, future studies should explore these issues. For more on this see Perrigo (2023).

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